

Concavity, Curve Sketching, and Optimization



Increasing/decreasing and the first derivative test

- 1 For $f(x) = x^3 - 6x^2 + 9x$, find the intervals where f is increasing and decreasing, and classify each critical point using the First Derivative Test.
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Concavity and the second derivative test

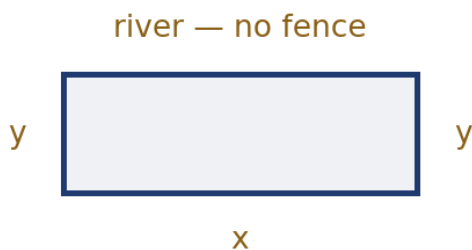
- 2 For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.
 - 3 Use the Second Derivative Test to classify the critical points of $f(x) = x^3 - 3x$.
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Curve sketching

- 4 For $f(x) = x^3 - 3x^2$, find: (a) critical points and local extrema, (b) inflection points, (c) intervals of increase/decrease and concavity. Summarize with a sign chart.
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Optimization problems

- 5 A farmer has 200 m of fencing to enclose a rectangular field bordering a river (no fence needed along the river). Find the dimensions that maximize the enclosed area.

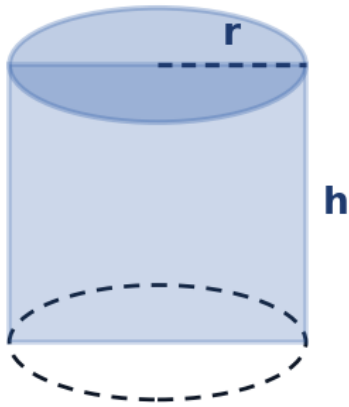


- 6 An open-top box is made from a 12 in by 12 in square sheet by cutting equal squares of side x from each corner and folding up the sides. Find x that maximizes the volume.
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Optimization: minimizing cost or distance

- 7 Find the point on the line $y = 2x - 3$ closest to the origin, by minimizing the squared distance $D = x^2 + y^2$.

- 8 A closed cylindrical can must have volume $16\pi \text{ cm}^3$. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi r h$, find the radius that minimizes surface area.



Points of inflection and graph behavior

- 9 Explain why a point of inflection does NOT necessarily correspond to a local maximum or minimum, using $f(x) = x^3$ at $x = 0$ as an example.

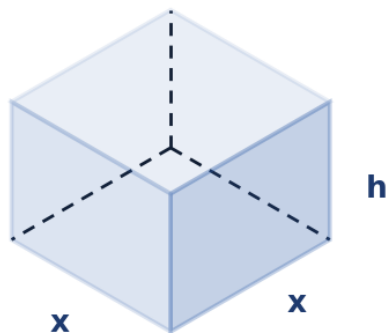
Visualizing extrema and concavity together

- 10 The graph shows $f(x) = x^3 - 3x^2$ from the curve-sketching problem above, with its local max at $x = 0$, local min at $x = 2$, and inflection point at $x = 1$.



A capstone optimization problem

- 11 A rectangular box with a square base and no top must have volume 500 cm^3 . Using $V = x^2h = 500$ (so $h = \frac{500}{x^2}$) and surface area $S = x^2 + 4xh$, find the base side length x that minimizes the surface area.



Second derivative test failures

- 12 For $f(x) = x^4$, show that the Second Derivative Test is inconclusive at the critical point $x = 0$, then use the First Derivative Test to correctly classify it.
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A capstone curve-sketching problem

- 13 For $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$, find all critical points and classify each using the Second Derivative Test.